



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Center

True Stereo Recentering

Pulls a drifting or off-center stereo source back to the middle, aligns its left and right arrival times, and keeps the width you want.

Direction tracking

Time alignment

M/S width

4-band option

Windows VST3 · macOS VST3/AU



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01 Overview

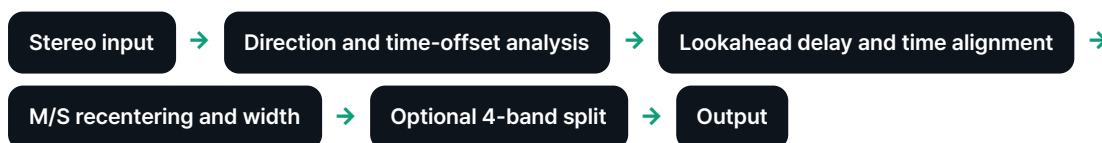
TB Center tracks where the dominant sound in a stereo signal sits between left and right, and rotates the stereo image so that sound returns to the center. It can also line up a source that reaches the two channels at slightly different times.

Use it on dialogue, vocals, or instruments that drift off-center, on stereo recordings where one source leans to one side, or on a stereo bus whose focus has moved. Unlike a simple balance or width control, TB Center follows the source as it moves and corrects only the direction it detects.

Key features

- Continuous tracking of the dominant source direction, with adjustable speed
- Time alignment that pulls left/right arrival differences back onto one point and reduces comb filtering in mono
- Lookahead analysis for steadier positioning
- Mid/Side width control after recentering, from mono to 200%
- Optional four-band mode with separate amounts below 200 Hz, 200 Hz–1 kHz, 1–5 kHz, and above 5 kHz
- Factory and user presets with undo and redo

Signal flow



About this manual

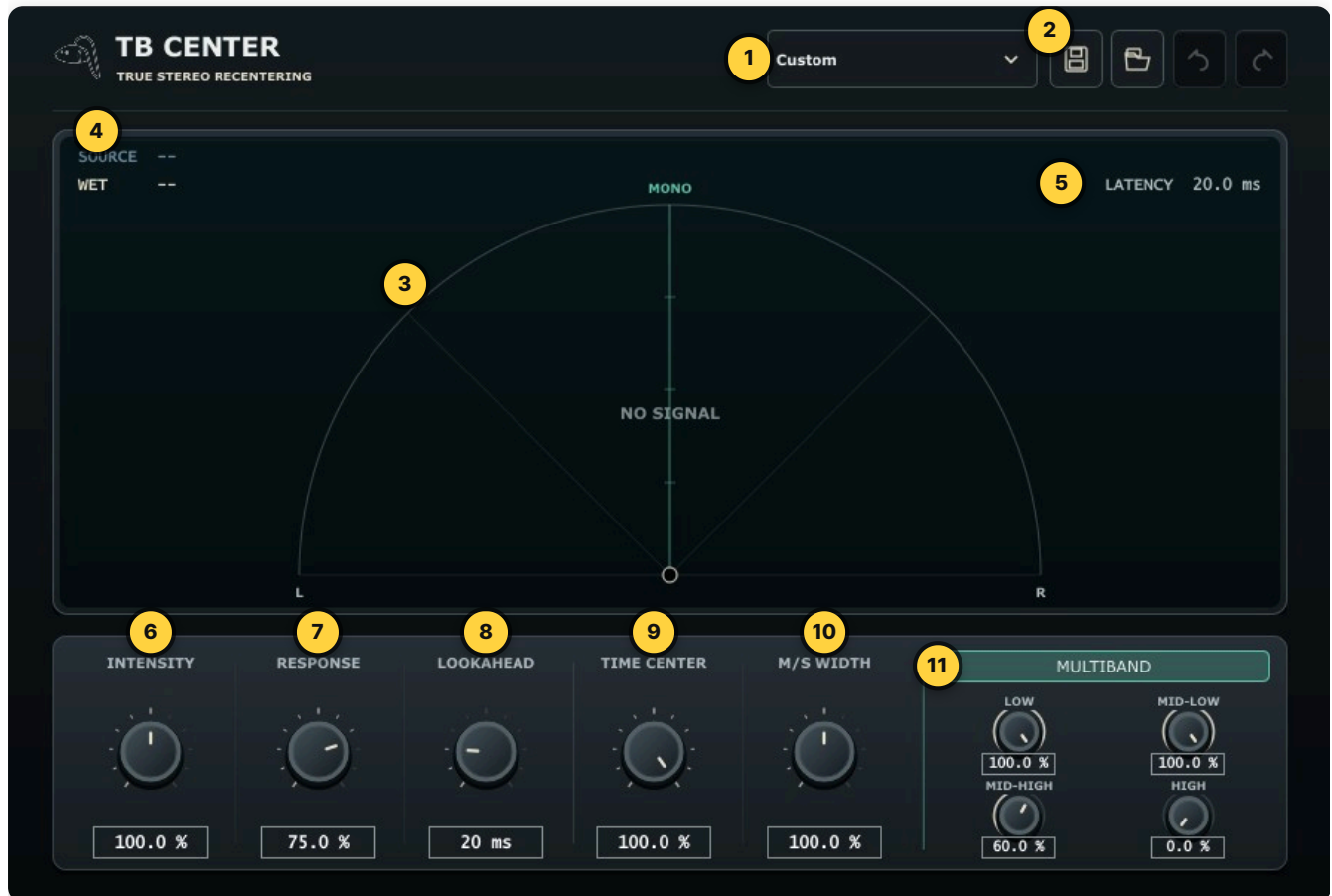
This manual describes TB Center 1.3.0 for Windows (VST3). The current macOS build is version 0.3.4 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Center on a stereo track. Start from the defaults: **INTENSITY** 100%, **RESPONSE** 75%, **LOOKAHEAD** 20 ms, **TIME CENTER** 100%, **M/S WIDTH** 100%.
- 2 Play the material and watch the direction display **3**. It shows where the dominant source sits between L and R.

- 3 Adjust **INTENSITY** 6 until the source holds the center without the rest of the image swinging with it.
- 4 Set **RESPONSE** 7: higher values follow a moving source quickly; lower values give a calmer, steadier image.
- 5 If the source sounds smeared or phasey in mono, raise **TIME CENTER** 9. If it sounds unnatural, lower it.
- 6 Set the final width with **M/S WIDTH** 10, then compare with bypass at the same level.

03 Interface



- 1 **Preset list** — Factory presets and your own presets.
- 2 **Save, Load, Undo, Redo** — save or load a .tbcenter preset file and step through recent changes.
- 3 **Direction display** — the dominant source between L and R; the vertical line marks the center (MONO). NO SIGNAL appears when there is no input.
- 4 **SOURCE and WET** — readouts for the detected source and the processing state. WET shows BYPASSED when the plug-in is bypassed.
- 5 **LATENCY** — the delay TB Center currently adds, set by LOOKAHEAD.
- 6 **INTENSITY** — how strongly the image is turned toward the center.
- 7 **RESPONSE** — how quickly the detected direction is followed.
- 8 **LOOKAHEAD** — analysis time ahead of the audio, for steadier positioning.
- 9 **TIME CENTER** — how much left/right arrival-time difference is corrected.
- 10 **M/S WIDTH** — the width of the recentered image.
- 11 **MULTIBAND** — switches on four-band processing with LOW, MID-LOW, MID-HIGH, and HIGH amounts.

04 Controls

Control	Range · default	What it does
INTENSITY	0 to 200% · 100%	Correction strength toward the center. It is not a dry/wet mix: 0% leaves the direction unchanged, 100% centers the detected source, and values above 100% overcorrect.
RESPONSE	0 to 100% · 75%	How quickly the detected source direction is followed. Raise it for sources that move; lower it if the image wobbles.
LOOKAHEAD	from 0 ms · 20 ms	Lets the analysis see slightly ahead for steadier positioning. It adds the same amount of latency, and a new value takes effect when you release the knob. It cannot be automated.
TIME CENTER	0 to 100% · 100%	Pulls a source that reaches L and R at different moments back onto one point, so a smeared image tightens and the mono sum stops comb filtering. 0% keeps the original arrival times.
M/S WIDTH	0 to 200% · 100%	Width of the centered image. 100% keeps it as centered, 0% collapses it to mono, and values above 100% widen it.
MULTIBAND	Off, On · Off	Off applies one correction to the whole signal. On splits the spectrum into four fixed bands and recenters each one independently.
LOW	0 to 100% · 100%	Recentering amount below 200 Hz (with MULTIBAND on).
MID-LOW	0 to 100% · 100%	Recentering amount from 200 Hz to 1 kHz.
MID-HIGH	0 to 100% · 100%	Recentering amount from 1 kHz to 5 kHz.
HIGH	0 to 100% · 100%	Recentering amount above 5 kHz. 0% leaves wide highs untouched.

With MULTIBAND on, keep LOW at 100% to hold the bass and kick in the middle, and lower HIGH to keep the natural width of cymbals, room, and reverb.

05 How it reacts

TB Center decides how far to turn the image by analyzing the signal, not by a fixed pan setting. It works best when one dominant source, such as a voice or a lead instrument, is clearly louder than the rest.

Time alignment depends on how similar the two channels are. On a single source captured by two microphones it can tighten the image noticeably; on uncorrelated material, such as wide reverbs or stereo pads, it has little to work with and changes little.

LOOKAHEAD adds latency, shown as LATENCY in the display. Keep it low when you monitor through the plug-in while recording.

06 Presets

Choose a preset from the list **1**. Save writes the current settings to a .tbcenter file, and Load opens one. Undo and Redo step back and forward through parameter and preset changes.

Factory presets: **Init**, **Locked Dialogue**, **Natural Center**, **Fast Motion**, **Mix Recenter**, and **Wide Keep Air**.

07 Recipes

Starting points for common jobs. Adjust them by ear.

Hold dialogue in the middle

A voice that drifts as the speaker moves or the boom swings.

Preset	Locked Dialogue
RESPONSE	Raise if the voice moves quickly
M/S WIDTH	Below 100% for a focused voice

- 1 Watch the direction display and raise INTENSITY until the voice stays centered.
- 2 Check that room tone does not swing with the voice.

Re-center a stereo bus

A mix or stem whose focus leans to one side.

Preset	Mix Recenter
MULTIBAND	On
LOW	100%
HIGH	Lower to keep the air wide

- 1 Use small INTENSITY values first; a bus has many sources.
- 2 Compare in mono for comb filtering.

Fix timing between two mics

A source recorded with two microphones at different distances.

TIME CENTER	100%
LOOKAHEAD	Default 20 ms

- 1 Switch your monitoring to mono and listen for a fuller, less hollow sound.

Center without losing width

Keeping the space of a wide recording while fixing its focus.

Preset	Wide Keep Air
M/S WIDTH	100% or more

- 1 Lower INTENSITY if the ambience starts to collapse.

08 Tips and troubleshooting

Symptom	What to check
The whole image swings around.	Lower RESPONSE, or lower INTENSITY. A busy mix may not have one clear dominant source.
The source sounds hollow in mono.	Raise TIME CENTER so left and right arrival times line up.
The sound became narrow.	Raise M/S WIDTH, or use MULTIBAND and lower HIGH to keep high-frequency width.
LOOKAHEAD does not change while I turn it.	It applies when you release the knob, and it cannot be automated.

Symptom	What to check
The plug-in adds delay.	That is the LOOKAHEAD latency shown at the top right of the display.
The display says NO SIGNAL.	No audio is reaching the plug-in. Check the track routing and input.

09 Parameter reference

Panel label	Host parameter	Range	Default
INTENSITY	Intensity	0 to 200%	100%
RESPONSE	Response	0 to 100%	75%
LOOKAHEAD	Not automatable	from 0 ms	20 ms
TIME CENTER	Time Center	0 to 100%	100%
M/S WIDTH	M/S Width	0 to 200%	100%
MULTIBAND	Multiband	Off, On	Off
LOW	Band Low	0 to 100%	100%
MID-LOW	Band Mid-Low	0 to 100%	100%
MID-HIGH	Band Mid-High	0 to 100%	100%
HIGH	Band High	0 to 100%	100%

10 Version history

Version	Changes
1.3.0 · 2026-07-30	The plug-in logo and name share one rack header. Previous/next navigation and save/load in the preset control were improved.

11 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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